CodeAlpha C Programming Internship

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Task 2: Medium Matrix Operations

Aim:
To write a program to implement Matrix Addition, Matrix Multiplication and Transpose

Program:

```
#include <stdio.h>

#define MAX 10

void readMatrix(int mat[MAX][MAX], int rows, int cols, const char *label) {

   printf("\nEnter elements of matrix %s (%d x %d), row by row:\n", label, rows, cols);

   for (int i = 0; i < rows; i++) {

      for (int j = 0; j < cols; j++) {

         printf("  %s[%d][%d]: ", label, i, j);

         scanf("%d", &mat[i][j]);

      }

   }

}

void printMatrix(int mat[MAX][MAX], int rows, int cols, const char *label) {

   printf("\nMatrix %s:\n", label);

   for (int i = 0; i < rows; i++) {

      printf("  [ ");

      for (int j = 0; j < cols; j++) {

         printf("%4d ", mat[i][j]);

      }

      printf("]\n");
```

```
    }
}
void addMatrices(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int rows,
int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            result[i][j] = a[i][j] + b[i][j];
        }
    }
}
 void multiplyMatrices(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX],
                      int rowsA, int colsA, int colsB) {
    for (int i = 0; i < rowsA; i++) {
        for (int j = 0; j < colsB; j++) {
            result[i][j] = 0;
            for (int k = 0; k < colsA; k++) {
                result[i][j] += a[i][k] * b[k][j];
            }
        }
    }
}
void transposeMatrix(int mat[MAX][MAX], int result[MAX][MAX], int rows, int cols) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            result[j][i] = mat[i][j];
        }
    }
}
int main() {
```

```
    int matA[MAX][MAX], matB[MAX][MAX], result[MAX][MAX];
    int rowsA, colsA, rowsB, colsB;
    int choice;
    printf("Matrix Operations\n");
    printf("Select an operation:\n");
    printf("1. Matrix Addition\n");
    printf("2. Matrix Multiplication\n");
    printf("3. Transpose of a Matrix\n");
    printf("Enter choice (1-3): ");
    if (scanf("%d", &choice) != 1 || choice < 1 || choice > 3) {
        printf("Invalid choice. Please run the program again and select 1, 2, or 3.\n");
        return 1;
    }
    if (choice == 1) {
        printf("\nEnter number of rows and columns (same for both matrices): ");
        scanf("%d %d", &rowsA, &colsA);
        if (rowsA <= 0 || colsA <= 0 || rowsA > MAX || colsA > MAX) {
            printf("Dimensions must be between 1 and %d.\n", MAX);
            return 1;
        }
        readMatrix(matA, rowsA, colsA, "A");
        readMatrix(matB, rowsA, colsA, "B");
        addMatrices(matA, matB, result, rowsA, colsA);
        printMatrix(matA, rowsA, colsA, "A");
        printMatrix(matB, rowsA, colsA, "B");
        printMatrix(result, rowsA, colsA, "A + B");
    }
  else if (choice == 2) {
        printf("\nEnter rows and columns of Matrix A: ");
```

```
        scanf("%d %d", &rowsA, &colsA);
        printf("Enter rows and columns of Matrix B: ");
        scanf("%d %d", &rowsB, &colsB);

        if (rowsA <= 0 || colsA <= 0 || rowsB <= 0 || colsB <= 0 ||
            rowsA > MAX || colsA > MAX || rowsB > MAX || colsB > MAX) {
            printf("Dimensions must be between 1 and %d.\n", MAX);
            return 1;
        }
        if (colsA != rowsB) {
            printf("\nError: Multiplication not possible. ");
            printf("Columns of A (%d) must equal rows of B (%d).\n", colsA, rowsB);
            return 1;
        }
        readMatrix(matA, rowsA, colsA, "A");
        readMatrix(matB, rowsB, colsB, "B");
        multiplyMatrices(matA, matB, result, rowsA, colsA, colsB);
        printMatrix(matA, rowsA, colsA, "A");
        printMatrix(matB, rowsB, colsB, "B");
        printMatrix(result, rowsA, colsB, "A x B");
    }
    else if (choice == 3) {
        printf("\nEnter number of rows and columns of the matrix: ");
        scanf("%d %d", &rowsA, &colsA);
        if (rowsA <= 0 || colsA <= 0 || rowsA > MAX || colsA > MAX) {
            printf("Dimensions must be between 1 and %d.\n", MAX);
            return 1;
        }
        readMatrix(matA, rowsA, colsA, "A");
```

```
        transposeMatrix(matA, result, rowsA, colsA);

        printMatrix(matA, rowsA, colsA, "A");
        printMatrix(result, colsA, rowsA, "A Transpose");
    }
    return 0;}
```

Output:

```
        Matrix Operations
Select an operation:
  1. Matrix Addition
  2. Matrix Multiplication
  3. Transpose of a Matrix
Enter choice (1-3): 2

Enter rows and columns of Matrix A: 2 3
Enter rows and columns of Matrix B: 3 2

Enter elements of matrix A (2 x 3), row by row:
  A[0][0]: 1  A[0][1]: 2  A[0][2]: 3
  A[1][0]: 4  A[1][1]: 5  A[1][2]: 6

Enter elements of matrix B (3 x 2), row by row:
  B[0][0]: 7  B[0][1]: 8
  B[1][0]: 9  B[1][1]: 10
  B[2][0]: 11 B[2][1]: 12

Matrix A:
  [    1    2    3 ]
```

[4 5 6]

Matrix B:

[7 8]

[9 10]

[11 12]

Matrix A x B:

[58 64]

[139 154]